

Codebook Distribution Alignment during Manifold Warmup: Preliminary Evidence on SimVQ Initialization and the Need for K-Means

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Abstract

This note investigates two questions in progressive VQ training: why SimVQ may exhibit more severe token representation shrinkage than vanilla VQ under Deferred Quantization, and whether K-Means is necessary to bridge Manifold Warmup and latent discretization. Three preliminary observations emerge. First, SimVQ freezes its base codebook and updates only a linear projection layer; consequently, the initial geometry of the effective codebook may be governed primarily by the projection matrix. Replacing identity initialization with an analytical covariance-based initialization derived from the warmed-up latent space alleviates token representation shrinkage. Secondly, directly adding a codebook loss during vanilla VQ warmup substantially degrades performance, plausibly because codewords receive sparse, local updates. By contrast, an optimal-transport distribution-matching regularizer that updates the codebook globally aligns it with the latent space before quantization and performs better overall than the baseline. Thirdly, in Stage 2, SimVQ with random initialization remains comparable to its K-Means-initialized counterpart and is slightly better on several metrics, possibly because the shared projection matrix can update all codewords jointly.

1 Motivation and Hypotheses

EQSC identifies early quantization as a cause of token representation shrinkage and reports that, under Deferred Quantization, shrinkage is more pronounced for SimVQ than for vanilla VQ. SimVQ freezes a base codebook during training and optimizes only a linear projection layer. Thus, even when the base codebook is initialized by K-Means, the initial geometry of the effective codebook may be determined largely by the projection matrix. Identity initialization need not match either the scale or the correlation structure of the latent representations learned during warmup. Motivated by this observation, I introduce a covariance-based initialization derived from the statistics of the latent distribution. It whitens and recolours the base codebook so that the effective codebook matches the second-order moments of the latent space before quantization begins.

ProVQ mitigates the optimization deadlock caused by premature discretization by first learning a continuous latent manifold and then introducing the discretization constraint. The transition between these stages, however, normally relies on K-Means. I therefore examine whether the codebook can learn information about the latent space directly during Stage 1. Adding a standard codebook loss during warmup causes performance to deteriorate sharply. One possible explanation is that nearest-neighbour assignments provide local gradients only to the small subset of selected codewords, amplifying sparse updates and the rich-get-richer effect. Inspired by optimal transport (OT), I instead use stop-gradient Sinkhorn distribution matching: the encoder continues to be updated solely by the continuous autoencoder objective, while gradients from the matching objective flow only to the codebook. This replaces pointwise nearest-neighbour regression with global distribution matching.

2 Experimental Setup

All experiments use CIFAR-10 with images normalized to $[-1, 1]$. The encoder and decoder architectures are held fixed, the latent dimension is 32, and the codebook contains 8,192 entries. Models are trained for 100 epochs with Adam, a learning rate of 2×10^{-4} , and a batch size of 128. The first 50 epochs constitute the autoencoder Manifold Warmup, followed by the hard-quantization stage. Evaluation metrics include PSNR, SSIM, perplexity (PPL), codebook utilization (Usage), and average pairwise Euclidean distance between codewords (Euc.). All results are reported as the mean and standard deviation over multiple runs.

The principal comparisons include standard VQ-VAE, K-Means initialization, Manifold Warmup followed by either random or K-Means initialization, and the corresponding SimVQ variants. The ablations comprise covariance-based initialization (Cov Init), a codebook loss during warmup (CL Warmup), and a codebook-only Sinkhorn regularizer. For the last method, I compare disabling the regularizer in Stage 2 with annealing its weighting coefficient.

3 Preliminary Results

Base Model	Variant	PSNR	SSIM	PPL	Usage (%)	Euc.
Vanilla	Baseline	24.79±0.11	0.847±0.004	719±77	10.86±1.24	1.87±0.05
	+ K-Means	24.39±0.22	0.834±0.007	386±98	5.53±1.45	0.48±0.07
	+ Warmup	25.07±0.02	0.859±0.001	1092±63	18.25±0.99	2.00±0.05
	+ Warmup + K-Means	25.00±0.02	0.860±0.001	4373±441	69.68±6.11	6.96 ±0.90
	+ CL Warmup	24.08±0.78	0.825±0.027	416±324	6.82±6.59	1.70±0.25
	+ Sinkhorn	25.12±0.03	0.863±0.002	5161±578	78.77±7.79	6.65±1.06
	+ Sinkhorn*	25.10±0.30	0.869±0.005	6902 ±235	96.51±2.28	6.34±1.02
SimVQ	+ Warmup	25.70 ±0.06	0.871±0.001	4652±113	99.90±0.05	5.25±0.08
	+ Warmup + K-Means	25.58±0.40	0.875 ±0.002	6210±156	99.91 ±0.02	5.97±0.12
	+ Warmup + Cov Init	25.52±0.19	0.867±0.006	4524±494	98.75±1.06	6.75±1.41
	+ CL Warmup	25.65±0.10	0.869±0.003	4753±32	99.81±0.14	5.83±0.41

Table 1: Preliminary ablation results on CIFAR-10. The asterisk denotes Sinkhorn regularization with an annealed weighting coefficient.

The preliminary results support the following observations:

- Unlike SimVQ, vanilla VQ exhibits clear training degradation when CL Warmup is enabled. This suggests that local, pointwise codebook pre-adaptation does not extract useful guidance from the encoder and cannot replace K-Means. Sinkhorn pre-adaptation substantially improves this behaviour: the annealed variant attains the highest PPL, while its reconstruction quality exceeds that of the Warmup + K-Means baseline.
- One plausible explanation is that global distribution matching (OT methods represented by Sinkhorn) facilitates efficient decoupled learning. The codebook can acquire useful information from the encoder’s latent space before hard quantization, without interfering with continuous representation learning, thereby reducing the optimization burden during the hard-quantization stage.
- SimVQ appears less dependent on K-Means than vanilla VQ. With a randomly initialized base codebook, its PSNR is comparable to, and slightly higher than, that of the K-Means variant, while both achieve nearly 100% utilization. Relative to K-Means initialization, covariance-based initialization preserves high utilization and moves the codeword-distance metric closer to the level observed for vanilla VQ. This provides preliminary evidence that the initialization of the linear projection layer affects the geometric evolution of the effective codebook.

4 Conclusion and Next Steps

The preliminary evidence suggests that the essential bridge between Manifold Warmup and latent discretization is not K-Means itself, but effective coordination and alignment between the latent space and the codebook before quantization. Because SimVQ can move all codewords jointly through a shared projection layer, it is relatively robust to the initialization of the base codebook. Vanilla VQ, by contrast, benefits from avoiding sparse pointwise pre-adaptation. Global OT-based distribution matching is a promising alternative; it also motivates an explicit three-stage training scheme in which OT alignment serves as an intermediate transition between the two original stages.

Future work will include more random seeds and datasets, track changes in the effective rank of the latent space and the geometry of the codebook around the quantization transition, and compare alternative OT training schemes to determine whether these findings generalize reliably.